

Operant Bandit Task with Average-Reward Reinforcement Learning

Methods, analyses, and results extracted from the accompanying study

Abstract

This document isolates the operant bandit component of the accompanying study of average-reward reinforcement learning (RL), recurrent neural networks (RNNs), and ventral-striatal dopamine signals. It collects the bandit task specification, model inputs and actions, actor–critic training procedure, analysis definitions, training parameters, behavioral and RPE results, population-dynamics results, and the three figures associated with the bandit task. Conditioning-task-specific results are omitted.

1 Operant bandit task

The behavioral task was described in previous work (Mohebi et al., 2019, 2024). A trial started when the center-port light turned on (light-on). After the rat poked into the center port (center-in), it held there for a variable period of 0.5–1.5 s until a go-cue occurred and the lights at the two adjacent ports turned on. The rat then freely chose one of the two adjacent ports (side-in), each with a predefined probability of triggering a click indicating reward delivery at the food port. The reward probabilities for the two side ports included all combinations of 0.1, 0.5, and 0.9 and remained constant during blocks of 40–60 trials. After a block ended, the reward probabilities changed randomly without notification. Inter-trial intervals ranged from 5–10 s.

In the model, the action space included “non-engaged”, “engage”, poking into each of the three ports, poking into the food port, and “movement”. During the inter-trial interval, the action started with “non-engaged”, then switched to “engage” before poking into the middle port (center-in) and starting the holding period. Transition from one action to another required passing through at least one “movement” action.

The RNN input contained 16 dimensions. Environmental signals consisted of light-on signals for the three ports (3 dimensions), the go-cue (1), reward click (1), and background input (3). The remaining inputs were the reward received at the previous time step (1) and a one-hot representation of the previous action (7, equal to the action-space size).

Dopamine signals reported in the associated experimental work were measured using fiber photometry of the fluorescent sensor dLight in rat ventral striatum. For the bandit task, recordings began after the rats had fully learned the task (Mohebi et al., 2019, 2024).

2 RNN actor–critic model and training

We used an actor–critic RNN composed of LSTM units (Hochreiter and Schmidhuber, 1997; Mnih et al., 2016; Wang et al., 2018). The loss contained policy, value, and entropy terms:

$$L^\theta = L^P(\theta) + \beta_V L^V(\theta) - \beta_E L^E(\theta). \quad (1)$$

The network was trained with APO (Ma et al., 2021), an average-reward generalization of proximal policy optimization (PPO) (Schulman et al., 2017). During training, average reward was updated as

$$\hat{\eta} \leftarrow (1 - \alpha)\hat{\eta} + \alpha \frac{1}{T} \sum_{t=0}^{T-1} r_t, \quad (2)$$

where r_t is reward at time step t and T is sequence length.

The policy loss was

$$L_t^P(\theta) = \min\left(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t\right), \quad (3)$$

where $\rho_t = \pi_\theta(a_t|s_t)/\pi_{\theta_{old}}(a_t|s_t)$ and $\hat{A}_t$ was the generalized advantage estimator (GAE) (Schulman et al., 2015) of

$$\delta_t = r_{t+1} - \hat{\eta} + \hat{V}_{t+1} - \hat{V}_t. \quad (4)$$

The value loss was

$$L_t^V(\theta) = \frac{1}{2}(\bar{r}_t - \nu b - V_t)^2, \quad (5)$$

where $b = T^{-1} \sum_{t=0}^{T-1} \hat{V}_t$. For the bandit task, $\bar{r}_t = \hat{A}_t + \hat{V}_t$ was used to better account for the influence of GAE. The entropy term encouraged exploration (Mnih et al., 2016), and weights were updated with Adam (Kingma and Ba, 2014).

The bandit network was trained with 64 parallel environments for sampling efficiency and a sequence length of 40 s. Network-activity analyses considered the trained network after choice behavior had been optimized. The implementation was based on CleanRL (Huang et al., 2022) and extended to average-reward RL following Ma et al. (2021). Simulations were performed on a CPU cluster.

Before learning the full task, the network was trained sequentially to learn center-in, side-in, and food-port-in actions, receiving rewards after learning each procedure. This procedural pretraining resembled the sequential steps used to train rats on the task.

Table 1: Network and training parameters for the bandit task.

Parameter	Bandit task
Time step dt	0.1 s
Sequence length T	40 s
LSTM units	64
Input dimension	16
Action space	7
Action cost	0
Learning rate	0.0002
GAE λ	0.95
PPO clip ϵ	0.1
β_V	0.5
β_E	0.05
α	0.1
ν	0.5

3 Bandit-specific analyses

The RPE compared with dopamine was defined as

$$\text{RPE}_t = r_t - \hat{\eta} + \hat{V}_t - \hat{V}_{t-1}. \quad (6)$$

For the bandit task, reward rate ρ was defined with a leaky integrator,

$$\rho_t = (1 - \alpha_0)\rho_{t-1} + \alpha_0 r_t, \quad \alpha_0 = 0.001. \quad (7)$$

Evaluation trials from 200 blocks, excluding the first block, were divided into three quantiles (High, Med, and Low) according to reward rate at light-on.

4 Behavioral adaptation and RPEs

When trained using average-reward RL, the model displayed adaptive choice behavior (Fig. 2) and reached an average trial-wise reward of 0.61, above the stated chance level of 0.5; the maximum for an agent with complete knowledge of the

reward probabilities was 0.67. At the reward cue or omission, which occurred at side-in, RPEs scaled inversely with recent reward rate (Fig. 1a and Fig. 3d), resembling observations in rats trained on the same task (Mohebi et al., 2019, 2024). Trial onset (light-on) also evoked smaller RPEs (Fig. 3a); these scaled positively with reward rate, reflecting greater expectation of upcoming reward when more recent trials had been rewarded.

5 Population dynamics

Unit activity immediately before side-in, displayed in the first two PCA axes, clustered according to both reward history (Fig. 1b, right) and left/right choices (Fig. 1c, right). This separation was particularly clear across trial-block types. Activity at side-in for blocks with high, medium, and low reward rates occupied different regimes along PC1 (Fig. 1b, left). In four block types with clearly different left-versus-right reward probabilities, activity clustered along PC2 (Fig. 1c, left).

This clustering was already apparent before trial start at light-on (Fig. 3b,c). Thus, the network occupied dynamical regimes reflecting recent prior experience and adaptively biased the choice on the upcoming trial, consistent with a meta-learning interpretation (Wang et al., 2018).

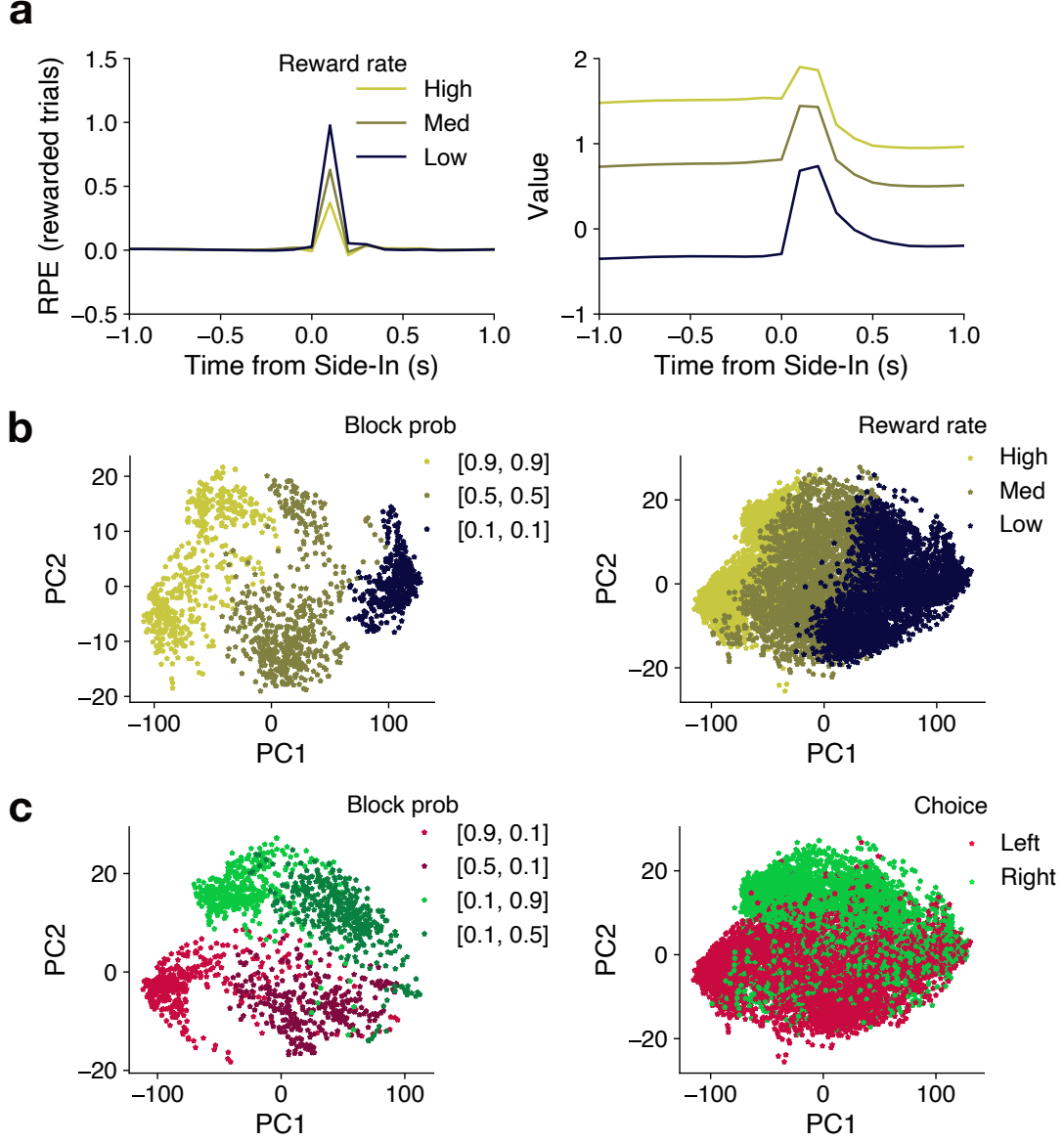


Figure 1: RPEs, values, and population activity for the RNN model trained using average-reward RL in the bandit task. **a.** RPEs (left) and values (right) at the side-in event. **b.** Clustering of population activity at side-in reflecting reward history. Left, data from the last 20 trials in blocks with high ([0.9, 0.9]), medium ([0.5, 0.5]), and low ([0.1, 0.1]) reward probabilities. Right, all trials with high, medium, and low reward rates. **c.** Clustering of population activity at side-in reflecting choice bias. Left, data from the last 20 trials in each of the four block types in which one choice is better than the other, including left-better blocks ([0.9, 0.1], [0.5, 0.1]) and right-better blocks ([0.1, 0.9], [0.1, 0.5]). Right, all trials labeled with choice in the current trial. Data were from 200 blocks of evaluation.

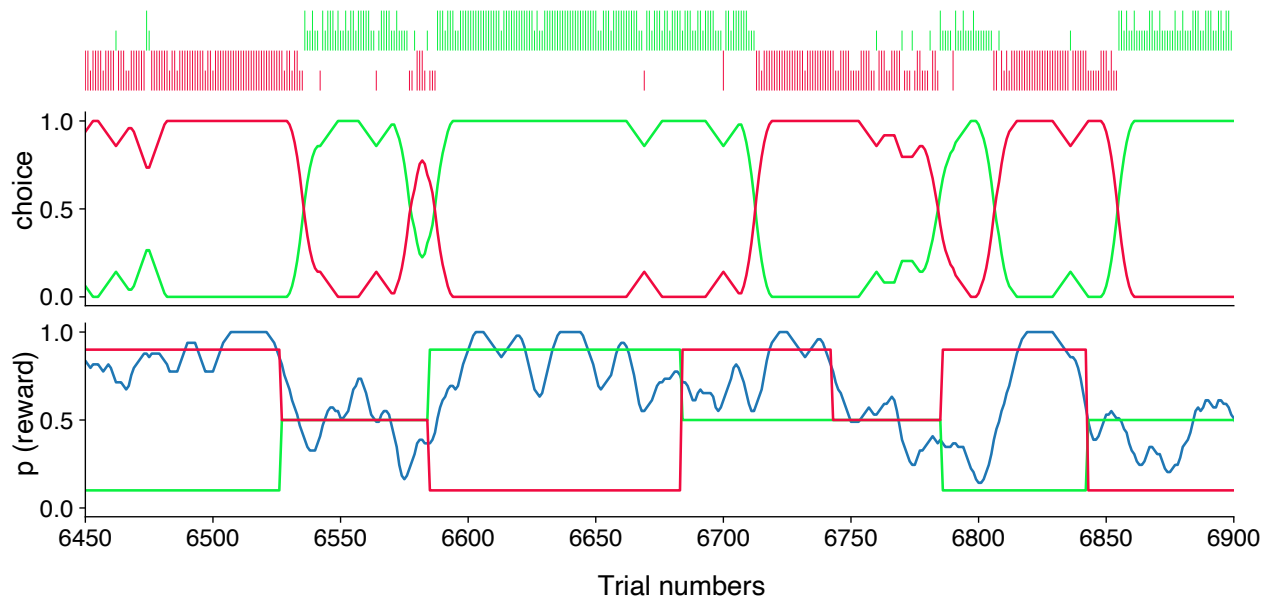


Figure 2: Example of behavioral choice adaptation from the RNN model trained with average-reward RL in the bandit task. Top, rewarded (long bars) and non-rewarded (short bars) trials for left (red) and right (green) choices. Middle and bottom panels show choosing probabilities and reward probabilities, respectively.

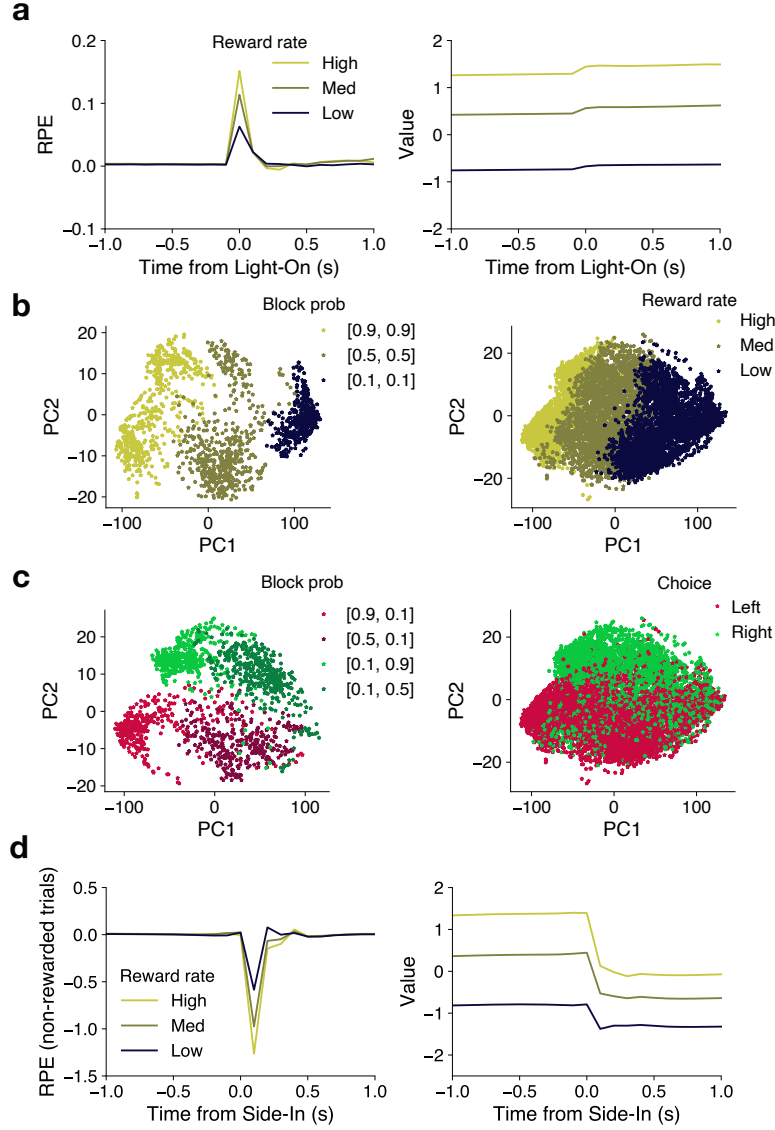


Figure 3: Additional results for RPEs, values, and unit activities from the RNN model trained using average-reward RL in the bandit task. **a.** Dependence of RPEs (left) and values (right) on reward rate when aligned with the light-on event. **b.** Clustering of population activities at light-on reflecting reward history. Left, the last 20 trials in blocks with high ([0.9, 0.9]), medium ([0.5, 0.5]), and low ([0.1, 0.1]) reward probabilities. Right, all trials with high, medium, and low reward rates. **c.** Clustering of population activities at light-on reflecting choice preference. Left, the last 20 trials in four block types with clearly distinct reward probabilities, including left-better blocks ([0.9, 0.1], [0.5, 0.1]) and right-better blocks ([0.1, 0.9], [0.1, 0.5]). Right, population-activity states for all trials, labeled by choice in the current trial. **d.** Dependence of RPEs (left) and values (right) on reward rate for non-rewarded trials when aligned with the side-in event. Data were from 200 blocks of evaluation.

6 Summary

The operant bandit results show that an RNN trained with average-reward RL can adapt its choices to changing reward probabilities while producing RPEs that depend systematically on recent reward rate. Its recurrent population state also carries information about reward history and upcoming choice before the choice event itself. In the parent study, these results support the broader examination of average-reward RL as a model of long-timescale reward prediction and ventral-striatal dopamine dynamics.

References

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